Telescoping Sonar Arm

The physical design reported for the arm uses a Teensy 4.1, 12 V BRINGSMART JGY-370 worm gearmotor, HiLetgo BTS7960 H-bridge, E38S6-600-24G incremental encoder, and one full-retraction mechanical limit switch. Measured length is about 0.846 m retracted and 1.380 m extended.

This is not yet a commissioned runtime contract. The checked-in `hardware.yaml` configuration is disabled, contains placeholder geometry, and represents the reported one-switch design: both NC and NO contacts of the minimum SPDT switch are monitored, while the maximum-switch pins are unassigned. ORACLE correspondingly disables automatic telescope-length commands. Motor drive remains disabled until the switch, geometry, encoder, current sensing, and wiring are commissioned together. Once enabled in a future measured configuration, limit-contact disagreement, overcurrent, invalid calibration, homing timeout, and encoder stall latch motor shutdown until reboot or a future explicitly reviewed reset path. Ordinary command timeout and reaching an endpoint remain nonfatal stops.

Homing and position

In the proposed one-switch design, the retraction switch is the physical reference. A completed homing cycle retracts slowly until that switch activates, removes motor drive, and sets the encoder position to zero. The present firmware does not unconditionally home at startup: it homes only after a minimum-length request unless it already observes the minimum switch. Automatic positioning remains unavailable until the hardware contract, switch logic, geometry, and homing behavior are commissioned together.

A one-switch implementation can omit a full-extension switch when packaging makes it physically impractical. Its extended end relies on a calibrated maximum encoder count, slow-down region, motor timeout, lack-of-motion detection, and a mechanical hard stop as the final passive boundary. Repeated loaded contact with the stop is not normal control behavior. Firmware supports an optional maximum SPDT switch, but the checked-in contract does not invent one.

The encoder measures a constant-radius capstan rather than the changing-radius storage spool. Runtime position uses a measured full-travel encoder calibration, not an unused motor-revolution proxy.

Current and packaging

The present firmware, configured through `firmware_config.h`, reads BTS7960 current-sense inputs, but the hardware configuration still marks current feedback unavailable and no calibrated motor-current contract exists. If those driver outputs cannot provide adequate bidirectional accuracy and range, an external Hall-effect sensor belongs in the motor supply path after the 12 V branch fuse and before the H-bridge. Either method must cover startup and stall current. Current supports jam/overload detection only after offset, polarity, normal-motion, startup, and stall-related thresholds are measured.

High-current motor wiring stays short and separated from encoder and limit signals. The home switch uses normally-closed logic where practical so a broken wire fails as a fault. Connectors must preserve current rating, locking, serviceability, contact protection, and strain relief.

The BTS7960 inputs and enable lines require external hard pulldowns plus a hardware driver-disable/interlock. Reset, boot, a disconnected MCU, or high-impedance GPIO must leave both bridge directions off independently of firmware.

If the main enclosure cannot provide volume for MCU, H-bridge, current sensor, fusing, regulators, LiDAR circuits, bend radii, and service loops, a small sealed arm-controller enclosure is safer and more serviceable than inaccessible stacking.

The current ROS surface in `runtime_surface.yaml` publishes length, motor current, and a status string. Homed state, extension percentage, direction, PWM, individual switch states, faults, and operating mode remain desirable structured fields, not current published contract fields.